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Team 6
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Individual Progress

For this week's system demo #1, we had five goals to demonstrate: CAD design, power system diagram, finalized list of motors, pseudo code for control architecture, and an updated gantt chart. Individually, I worked on the detailed CAD model for the full system design. First, I determined the size of the hexapod's base using the specs from the project requirements. The whole hexapod needs to fit within a 5 ft x 5 ft area, and needs to fit within a 3 ft x 3ft box for transportation after being deconstructed. Thus I designed the base to be 15 in. x 32 in. so that it is wide enough to sit on, and long enough to fit 1 person on each side. There are leg rests on either side for the riders to use. I also created a design for our 3 DOF legs. They are split into 3 segments: coxa, femur, and tibia. Combining these joints of different lengths allows for more rotation and articulation. I added cut-outs on each segment to reduce the weight of each piece to reduce the amount of torque needed to move them. The legs are also hollow with segments connecting both sides to allow space for wiring while still keeping the stability. Additionally, there is a rolling chair base supporting the main body of the hexapod to take care of the majority of the weight.

Aside from the CAD model, I also added to the updated gantt chart to include the estimated timeline for mechanical and electrical design, as well as the inverse kinematics and gait generation.

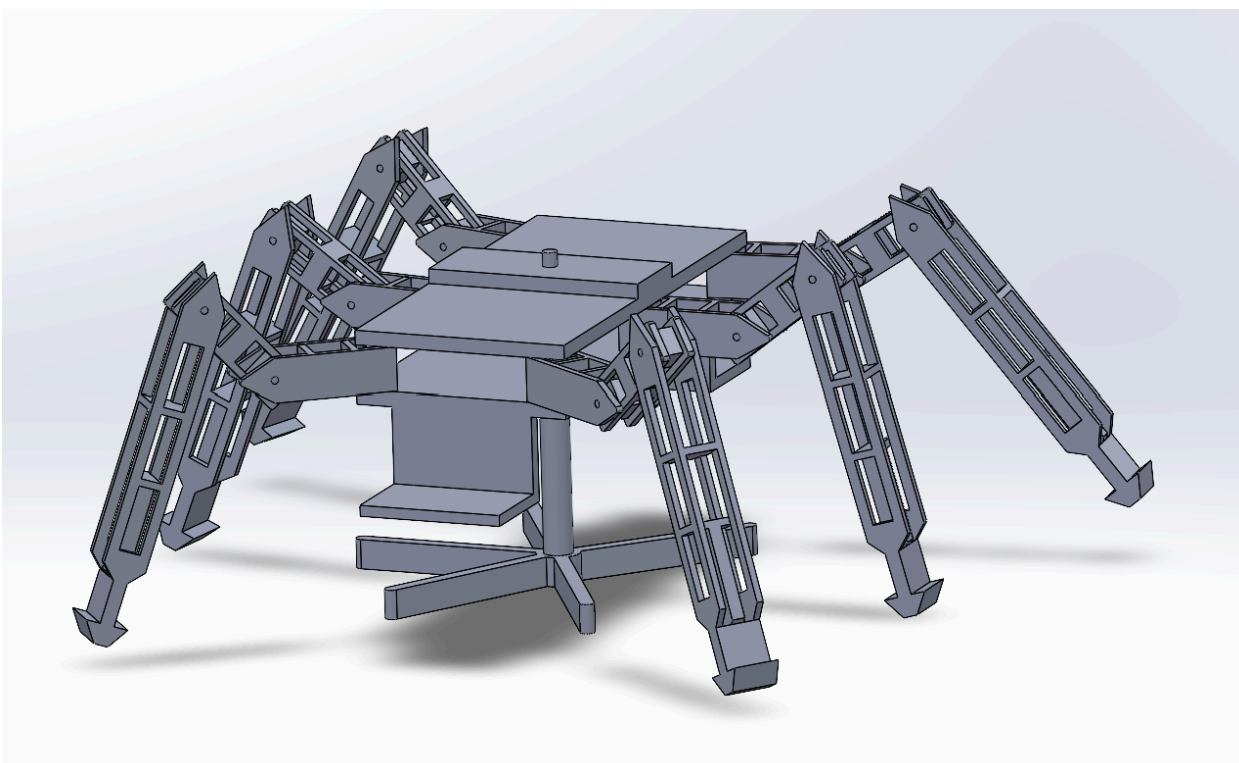

Figure 1: CAD model of full hexapod design

Challenges

The main challenge for creating this design was doing calculations to make sure that our leg weights were easily supportable by the motors that Mithun has considered. I also am

considering recalculating the leg lengths to make sure the height is sufficient for the 2-3 ft operating height. Additionally we need to consider the gait and the arc it will take to move. This will influence both the final design of the foot on the tibia joint, as well as the length of the legs to ensure a smooth movement.

Teamwork

For teamwork this week, we split the tasks up based on our strengths. I collaborated with Mithun to work on calculations for the hexapod base, as well as estimates for the leg joint lengths. I am also planning to continue working with Mithun on the gait generation, as well as the prototype for next week.

Plans

For next week, we have defined our goals for the next system demo. I am mainly focused on having a prototype of the legs to demonstrate motion. I also plan to have the leg lengths confirmed by the system demo. The other goals we have for system demo #2 are: power system demo, identification of final motor, have a physical mockup of the control architecture, and an update to our gantt chart.